

Notes: Giannetti, Liao & Yu (JF, 2015)
The Brain Gain of Corporate Boards: Evidence from China

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Data and measurement	4
3.1	Director biography data	4
3.2	Firm sample and performance outcomes	4
3.3	Provincial policies to attract highly skilled emigrants	4
3.4	Director-labor-market locality	5
3.5	Mechanism variables	5
4	Models and empirical specifications	5
4.1	First-stage director-supply equation	5
4.2	Baseline performance regression	5
4.3	Instrumental-variable model	6
4.4	Mechanism regressions	6
5	Identification and empirical design	6
5.1	Core identification problem	6
5.2	Policy timing as supply variation	6
5.3	Why locality matters	6
5.4	Threats and responses	7
6	Section-by-section study guide	7
7	Tables	7
7.1	Table I: Policies to Attract Highly Skilled Emigrants	7
7.2	Table II: The Geography of Directors Labor Market	8
7.3	Table III: Summary Statistics	9
7.4	Table IV: Policy Changes and the Board of Directors	10
7.5	Table V: Directors with Foreign Experience and Firm Value	11
7.6	Table VI: Directors with Foreign Experience and Total Factor Productivity	11
7.7	Table VII: Directors with Foreign Experience and Profitability	12
7.8	Table VIII: Powerful Firms and Directors with Foreign Experience	12
7.9	Table IX: Proportion of Directors with Foreign Experience and Board Characteristics	13
7.10	Table X: Business Education and Other Foreign Experience	14
7.11	Table XI: Internationalization	14

7.12 Table XII: Corporate Governance	16
7.13 Table XIII: Management Practices	16
7.14 Table XIV: Corporate Governance, Management Practices, and Corporate Valuations	17
8 Result map	18
9 Short summary	19
10 Conclusion	19
10.1 Core conclusion	19
10.2 Economic intuition	19
10.3 Incremental contribution revisited	19
10.4 Why the evidence is persuasive	19
10.5 Limitations	20
10.6 Takeaway	20

1 Big picture

- **Research question.** Does bringing directors with foreign education or foreign work experience onto corporate boards improve firm performance in an emerging market?
- **Core empirical setting.** The paper studies Chinese publicly listed nonfinancial firms from 1999 to 2009, a period in which Chinese provinces adopted policies at different times to attract highly skilled overseas Chinese emigrants.
- **Key treatment.** The main treatment is the proportion of a firm’s directors with foreign experience, where foreign experience includes foreign education, foreign work experience, or both.
- **Main identification idea.** Provincial policies shifted the local supply of potential directors with foreign experience. Because the director labor market is largely local, firms headquartered in treated provinces had more access to such directors after policy adoption.
- **Main outcome results.** Firms with more directors with foreign experience have higher valuation, higher total factor productivity, and later higher profitability.
- **Main channel results.** Improvements are linked to board-level decisions: more international mergers and acquisitions, greater use of foreign investors in private placements, more foreign sales, lower earnings management, stronger pay-performance governance, and better operating efficiency.
- **Main interpretation.** Returnee directors do not merely add credentials. They transmit organizational knowledge, foreign market connections, and governance practices to firms that operate in an institutional environment where managerial and governance expertise is scarce.
- **Economic takeaway.** Brain drain can become brain gain when emigrants return or reconnect with domestic firms through governance roles. The board of directors is a concrete mechanism through which international human capital enters emerging-market firms.

2 Incremental contribution of the paper

- **New board-capital channel.** The paper shifts the brain-drain/brain-gain discussion from workers, entrepreneurs, or foreign owners to the board of directors as a vehicle for transmitting managerial knowledge and governance practices.
- **First evidence on foreign-experienced returnees on boards.** The paper is among the first to study whether foreign education and foreign work experience of board members affect emerging-market firm outcomes.
- **Hand-collected director biography data.** The authors construct a director-firm-year database from biographies of tens of thousands of directors, coding foreign education, foreign work experience, nationality, political connections, and other director characteristics.
- **Quasi-experimental supply shock.** Instead of comparing firms that voluntarily hire foreign-experienced directors to firms that do not, the paper exploits staggered provincial policies to attract skilled emigrants. This creates plausibly exogenous local variation in director supply.
- **Local director-labor-market evidence.** Table II documents that directors are geographically local. This is crucial because a provincial policy shock only predicts board composition if firms mostly recruit directors from nearby local labor markets.
- **Emerging-market governance contribution.** Much of the board literature studies the United States and Europe. This paper shows that boards can matter in China, where weak governance and talent shortages make external expertise especially valuable.
- **Mechanism decomposition.** The paper goes beyond firm value. It examines internationalization, governance, management practices, and the geography and type of directors’ foreign experience.

- **Distinguishing returnee experience from foreign ownership.** The paper highlights a governance channel that can work even where foreign ownership is limited by policy or institutions.
- **Increment relative to consulting/FDI spillover studies.** Prior work emphasizes foreign ownership or outside consultants. This paper shows that directors with decision rights and monitoring authority can create firm-level knowledge transfer.

3 Data and measurement

3.1 Director biography data

- The authors hand-collect biographical information for directors of Chinese A-share listed firms.
- The director data identify whether a director has foreign work experience, foreign education, foreign visiting-scholar/training/postdoc experience, or foreign degrees at bachelor, master, or doctoral levels.
- The data also contain director gender, age, tenure, independence, political connection, nationality, and busyness.
- The main director measure is *Foreign Experience*, usually aggregated to the firm-year level as the proportion of board members with foreign experience.

Detailed interpretation: The director data allow the paper to identify not only whether the board has international human capital, but also what type of foreign experience the director has and where it was acquired.

3.2 Firm sample and performance outcomes

- The firm sample consists of nonfinancial Chinese listed companies from 1999 to 2009.
- Core performance outcomes are market-to-book ratio (MTB), total factor productivity (TFP), and return on equity (ROE).
- Outcomes are generally adjusted by subtracting industry-year medians, so the coefficients capture relative performance within industry-year cells.
- Control variables include ownership structure, state ownership, blockholders, foreign ownership, size, leverage, business segments, cash flow, young IPO status, volatility, and lagged performance.

Detailed interpretation: The outcome construction is designed to compare firms to contemporaneous peers in the same industry, reducing concerns that aggregate industry shocks drive the results.

3.3 Provincial policies to attract highly skilled emigrants

- Provinces implemented policies to encourage highly skilled overseas Chinese to return.
- The policies included incentives such as tax breaks, subsidized housing, tax-free imports, grants, medical benefits, schooling for children, spouse jobs, and residence permits.
- Policies were adopted at different times across provinces.
- The key instrument is a post-policy dummy for the province where the firm is headquartered, sometimes interacted with ownership variables.

Detailed interpretation: The policy timing creates variation in local director supply. The paper argues that adoption timing was not primarily driven by public-company board needs.

3.4 Director-labor-market locality

- The paper hand-collects the location of directors' primary employers.
- Table II shows that the majority of directors reside in the same province where the firms they serve are headquartered.
- This local labor-market evidence justifies using provincial policy changes as supply shocks for nearby firms.

Detailed interpretation: If director markets were national, local provincial policies would be much weaker instruments. Table II is therefore a key bridge between the institutional shock and firm-level board changes.

3.5 Mechanism variables

- **Internationalization:** foreign M&A, foreign private placements, and foreign sales.
- **Corporate governance:** earnings management, CEO turnover-performance sensitivity, CEO pay-performance sensitivity, and announcement returns around foreign acquisitions.
- **Management practices:** asset turnover and interactions with countries ranked highly in management-practice measures.
- **Ability controls:** rankings of Chinese universities attended by directors are used to proxy local ability.

Detailed interpretation: The paper does not stop at valuation effects. It asks whether the changes are consistent with specific board functions: advising, monitoring, and connecting firms to foreign opportunities.

4 Models and empirical specifications

4.1 First-stage director-supply equation

$$ForeignExperience_{i,p,t} = \alpha_i + \tau_t + \pi Policy_{p,t} + \Gamma X_{i,t} + u_{i,p,t}. \quad (1)$$

- $ForeignExperience_{i,p,t}$ is the proportion of directors with foreign experience in firm i headquartered in province p .
- $Policy_{p,t}$ equals one after province p adopts a policy to attract highly skilled emigrants.
- Some specifications interact the policy dummy with ownership variables measured before treatment.

Detailed interpretation: A positive and strong π means the policies shifted the local supply of directors with foreign experience. Table IV provides the key evidence.

4.2 Baseline performance regression

$$Y_{i,t+1} - \tilde{Y}_{industry,t+1} = \beta ForeignExperience_{i,t} + \Gamma X_{i,t} + \alpha + \tau_t + \epsilon_{i,t+1}. \quad (2)$$

- Y is MTB, TFP, or ROE.
- The dependent variable is typically the industry-year adjusted outcome.
- β captures the effect of foreign-experienced directors on future firm performance.

Detailed interpretation: The OLS specification measures associations. The IV specification uses provincial policies to isolate the component of foreign-experienced board representation driven by local supply shocks.

4.3 Instrumental-variable model

$$Y_{i,t+1} = \beta \widehat{ForeignExperience}_{i,t} + \Gamma X_{i,t} + FE + \epsilon_{i,t+1}, \quad (3)$$

where $\widehat{ForeignExperience}_{i,t}$ is predicted from provincial policy shocks.

Detailed interpretation: The IV estimate is a local average treatment effect for firms whose board composition responds to the policy-induced supply of foreign-experienced directors.

4.4 Mechanism regressions

$$PolicyOutcome_{i,t+1} = \beta ForeignExperience_{i,t} + \Gamma X_{i,t} + FE + \epsilon_{i,t+1}. \quad (4)$$

- Policy outcomes include international M&A, foreign private placements, foreign sales, earnings management, CEO turnover-performance sensitivity, CEO pay-performance sensitivity, and asset turnover.
- Some regressions include interactions with same-country experience, high corporate-governance rankings, or high management-practice rankings.

Detailed interpretation: These regressions ask whether board foreign experience changes decisions that boards plausibly influence, rather than merely predicting generic performance.

5 Identification and empirical design

5.1 Core identification problem

Firms do not randomly hire directors with foreign experience. Better firms, globally ambitious firms, or firms with superior governance may be more likely to attract such directors. The paper therefore cannot rely only on cross-sectional comparisons.

5.2 Policy timing as supply variation

- Provincial policies increased the availability of highly skilled returnees.
- The policies were not targeted at listed-company boards.
- Adoption timing varied across provinces and does not appear to track obvious public-firm growth opportunities.
- Because director markets are local, the supply shock is stronger for firms headquartered in treated provinces.

Detailed interpretation: Identification comes from comparing firms affected by policy-induced local supply shocks to similar firms in provinces not yet affected.

5.3 Why locality matters

The instrument would be weak or invalid if firms could easily recruit directors from any province. Table II shows that directors mostly reside in the province where they serve. This supports the claim that provincial policies changed the relevant local supply.

5.4 Threats and responses

- **Policy endogeneity:** Provinces might adopt policies when local firms are improving. The paper includes controls, province trends, province fixed effects, industry-year adjustments, and lagged performance checks.
- **Firm demand for foreign expertise:** Firms about to internationalize may lobby for policies. The policies were broad talent-attraction programs, not board-specific programs.
- **Director ability versus foreign experience:** The paper controls for Chinese university quality and board characteristics; it also shows country-specific channels.
- **Foreign ownership confounding:** Ownership controls and interactions show that effects are not simply due to foreign blockholders.
- **Board restructuring:** Table IX controls for board age, independence, tenure, busyness, political connection, and foreign nationality.

Detailed interpretation: The design is not a randomized experiment, but the combination of staggered policy shocks, local director-market evidence, controls, and mechanism tests makes the interpretation credible.

6 Section-by-section study guide

1. **Introduction and background.** Understand why Chinese firms face shortages of international managerial talent and weak governance, and why returnee policies generate useful variation.
2. **Data section.** Focus on how directors' foreign experience is coded and how firm outcomes are measured relative to industry-year benchmarks.
3. **First-stage section.** Table IV is essential: it shows that provincial policies predict the supply of foreign-experienced directors.
4. **Main performance section.** Tables V–VII establish the core effects on valuation, TFP, and profitability.
5. **Robustness and alternative explanations.** Tables VIII–X ask whether effects are driven by political power, board structure, foreign director nationality, or ability.
6. **Mechanism section.** Tables XI–XIV show internationalization, governance, management-practice, and valuation channels.

7 Tables

7.1 Table I: Policies to Attract Highly Skilled Emigrants

Read:

- Table I lists the year in which each province adopted policies to attract highly skilled emigrants.
- It also reports the number of firms and firm-year observations before and after adoption.
- The share of directors with foreign experience generally rises after policy adoption.
- The timing varies across provinces, creating staggered policy variation.

Detailed interpretation: This table defines the key policy shock. It is central because the paper uses policy timing to predict changes in board composition. The table also shows that the policy shock is not concentrated in only one or two large provinces.

Table II
The Geography of Directors Labor Market
This table reports the distribution of local directors. The first column presents the provinces in which sample firms are headquartered. The first row shows the provinces in which directors reside. The unit of observation is the director-firm-year.

	Beijing	Tianjin	Hebei	Inner Mongolia	Shanxi	Shanghai	Zhejiang	Shandong	Henan	Hubei	Guangdong	Guangxi	Yunnan	Tibet	Shaanxi	Gansu	Ningxia	Xinjiang
Beijing	6,150	118	40	44	9	74	26	47	88	97	100	9	35	25	73	31	72	41
Tianjin	212	1,086	3	0	4	24	12	8	61	14	0	2	4	0	12	4	0	20
Hebei	438	38	2,176	4	6	12	0	10	22	11	21	30	12	10	12	47	18	15
Shanxi	187	22	20	1,839	6	6	2	0	10	3	34	5	13	6	11	11	0	6
Inner Mongolia	228	14	0	11	1,355	14	6	11	46	5	6	0	0	1	13	6	6	10
Shanghai	318	20	5	1	7	4,062	44	37	103	26	24	13	9	1	43	11	16	1
Zhejiang	205	44	0	10	5	65	2,419	53	61	56	18	3	0	0	7	4	10	1
Shandong	206	2	4	7	8	56	8	2,181	54	8	44	17	0	13	12	3	2	5
Henan	780	48	11	7	25	57	31	20	11,343	149	96	49	119	6	32	38	78	16
Hubei	659	43	25	6	6	24	4	5	423	6,398	94	46	29	10	21	1	52	19
Guangdong	548	21	6	19	3	27	11	15	373	70	6,584	19	20	8	13	8	36	8
Guangxi	279	12	1	0	5	24	2	6	104	66	52	3,002	1	6	11	5	12	10
Yunnan	242	8	3	14	0	17	3	6	97	6	34	5	3,486	11	2	11	7	8
Tibet	171	4	5	12	0	1	3	8	106	5	19	0	17	1,625	1	11	15	11
Shaanxi	638	44	9	2	0	25	11	49	117	15	57	4	32	8	5,817	23	40	0
Gansu	276	14	28	3	1	10	5	7	68	14	17	10	6	9	18	2,176	20	20
Ningxia	390	3	16	7	3	28	6	4	116	45	39	11	12	29	37	32	4,752	44
Xinjiang	416	4	34	9	1	30	13	11	40	22	8	10	17	31	1	79	3,063	98
Beijing	1,113	54	21	12	6	117	39	11	298	99	123	67	100	91	108	56	162	121
Tianjin	131	12	15	8	0	8	17	3	27	24	11	5	11	5	12	31	15	11
Hebei	202	44	11	0	16	13	1	3	71	7	28	17	13	0	27	8	70	25
Shanxi	216	0	0	16	1	0	3	2	43	7	40	7	19	7	0	6	13	28
Inner Mongolia	287	42	18	3	2	49	10	13	94	49	28	4	18	9	70	9	43	36
Shanghai	160	0	2	0	4	1	5	6	35	6	0	1	6	5	40	8	0	4
Zhejiang	160	4	19	0	1	16	0	3	24	26	18	0	6	4	8	0	9	3
Shandong	53	8	0	5	0	0	7	0	8	8	0	6	0	1	6	0	0	4
Henan	107	3	3	0	5	4	3	28	7	22	3	3	7	28	5	5	39	0
Hubei	152	0	7	12	2	1	0	5	58	12	25	14	8	6	21	8	25	0
Guangdong	171	12	0	0	3	19	4	0	28	44	2	0	7	3	0	3	6	9
Guangxi	108	0	0	0	5	23	4	0	14	5	4	1	0	2	1	2	45	2
Yunnan	207	12	2	4	1	4	0	0	84	11	7	0	0	0	2	0	0	1
Tibet																		
Shaanxi																		
Gansu																		
Ningxia																		
Xinjiang																		

Table II: The Geography of Directors Labor Market

Economic message: The empirical design starts from a government attempt to attract talent; the corporate-governance effect arises because some of that talent enters the local board market.

Table I
Policies to Attract Highly Skilled Emigrants

This table reports the year of the policy adoption, the number of unique sample firms, and the proportion of directors with foreign experience for each province that implements a policy to attract highly talented emigrants. The sample period is 1999 to 2009. "Issuing Year" is the year the policy was adopted. "After" corresponds to observations after the issuing year. "Before" corresponds to observations before and during the issuing year. For "Percent of directors with foreign experience," the unit of analysis is firm-year-director.

Province	Issuing Year	No. of Unique Firms	No. of Firm-Year Observations			Percent of Directors with Foreign Experience	
			Total	Before	After	Before	After
Anhui	1994	55	424	0	424	0	7.03
Beijing	2000	104	803	85	718	7.05	10.73
Chongqing	2005	30	262	175	87	6.63	9.57
Fujian	2000	61	464	70	394	5.14	9.97
Gansu	2003	23	193	90	103	2	3.21
Guangdong	1999	210	1,576	112	1,464	7.85	13.61
Guangxi	2005	26	222	129	93	2.94	7.99
Guizhou	2003	18	157	85	72	3.34	3.63
Hainan	2001	21	207	52	155	9.49	9.4
Hebei	2001	38	342	72	270	1.08	5.98
Heilongjiang	2002	33	318	118	200	1.56	7.35
Henan	1992	42	328	0	328	0	5.08
Hubei	2002	65	612	204	408	5.12	8.84
Hunan	2001	53	422	96	326	2.5	10.81
Inner Mongolia	2001	23	213	49	164	2.94	5.78
Jiangsu	2004	121	894	362	532	7.2	7.75
Jiangxi	2003	27	236	85	151	6.35	5.78
Jilin	2001	38	342	88	254	4.35	7.47
Liaoning	1999	61	532	43	489	3.41	6.59

Table I, part 1

Table I—Continued

Province	Issuing Year	No. of Unique Firms	No. of Firm-Year Observations			Percent of Directors with Foreign Experience	
			Total	Before	After	Before	After
Ningxia	2003	11	114	49	65	6.48	3.42
Qinghai	1999	10	94	7	87	0	3.25
Shaanxi	1995	30	253	0	253	0	5.33
Shandong	2005	99	783	434	349	5.58	7.69
Shanghai	2005	168	1,468	912	556	9.79	11.91
Shanxi	2007	26	220	181	39	4.02	4.19
Sichuan	2005	77	668	408	260	5.06	7.69
Tianjin	2001	27	242	49	193	3.1	10.86
Tibet	N/A	9	78	N/A	N/A	N/A	N/A
Xinjiang	2003	32	263	97	166	5.08	4.78
Yunnan	2001	27	229	49	180	4.09	7.87
Zhejiang	2001	128	871	140	731	4.81	9.5

Table I, part 2

7.2 Table II: The Geography of Directors Labor Market

Read:

- The rows and columns show the provinces where directors reside and where firms are headquartered.
- The diagonal entries are large, indicating that directors often live in the same province as the firms they serve.
- Beijing, Shanghai, Guangdong, Jiangsu, Zhejiang, and other developed regions are especially important director-market centers.

Detailed interpretation: The local nature of the director labor market gives bite to provincial policies. If directors were perfectly mobile nationally, a policy in one province would not strongly shift board composition for local firms.

Economic message: Local board labor markets convert a province-level talent policy into a firm-level governance shock.

7.3 Table III: Summary Statistics

Read:

- Panel A describes the director-level sample. Directors with foreign experience are relatively rare, but not negligible.
- Panel B describes firm-level variables such as MTB, TFP, ROE, board size, state ownership, foreign ownership, leverage, and volatility.
- Panel C gives the industry distribution of sample firms.

Detailed interpretation: The table shows that the analysis has enough variation in foreign-experienced directors, firm performance, and ownership structure to study treatment effects and heterogeneity. It also reveals that China is a setting where foreign board experience is scarce enough to plausibly have high marginal value.

Economic message: Scarcity of foreign-experienced directors is exactly what makes the policy-induced supply shock economically meaningful.

The Brain Gain of Corporate Boards

1641

Table III
Summary Statistics

Panel A summarizes characteristics of the directors of our sample firms from 1999 to 2009. The unit of observation is the director-firm-year. "Director with Foreign Experience" is a dummy equal to one if a director has foreign education or foreign work experience. "Director with Foreign Work Experience" is a dummy equal to one if a director has foreign work experience. "Director with Foreign Education" is a dummy equal to one if a director has foreign education. "Foreign Visiting Scholar/Training/Postdoc" is a dummy equal to one if a director was a visiting scholar or post-doc, or did a short-term training program. "Foreign Bachelor Degree" is a dummy equal to one if a director holds a bachelor degree from a foreign country. "Foreign Master Degree" is a dummy equal to one if a director holds a master degree from a foreign country. "Foreign Doctoral Degree" is a dummy equal to one if a director holds a doctoral degree from a foreign country. "Director Age" is the difference between the current year and the birth year of the director. "Female Director" is a dummy equal to one if the director is female. "Director Tenure" is one plus the difference between the current year and the year when the individual joined the board of a given firm. "Non-independent Director" is a dummy equal to one if a director also receives a salary as an employee of the firm. "Foreign Director" is a dummy equal to one if a director is a foreign national. "Busy Director" is a dummy equal to one if a director sits on the boards of two or more publicly traded companies. "Politically Connected Director" is a dummy equal to one if a director is a current or former government bureaucrat. Panel B reports summary statistics for the sample firms between 1999 and 2009. The unit of observation is the firm-year. ROE is the firm's return on equity led by one year. All variable definitions are in the Appendix. Panel C reports the industry distribution of the sample firms. Statistics are based on firm-year observations. The 21 industries are based on the official industry classification of the China Securities Regulatory Commission.

Panel A: Director Characteristics				
	Mean	Median	Std. Dev.	No. of Obs.
Director with Foreign Experience	0.081	0	0.273	133,581
Director with Foreign Work Experience	0.032	0	0.176	133,581
Director with Foreign Education	0.063	0	0.244	133,581
Foreign Visiting Scholar/Training/Postdoc	0.029	0	0.168	133,581
Foreign Bachelor Degree	0.006	0	0.08	133,581
Foreign Master Degree	0.017	0	0.129	133,581
Foreign Doctoral Degree	0.011	0	0.103	133,581
Director Age	48.219	47	8.881	133,565
Female Director	0.097	0	0.296	133,581
Director Tenure	2.001	2	1.051	133,581
Non-independent Director	0.392	0	0.488	133,469
Foreign Director	0.004	0	0.062	133,581
Busy Director	0.162	0	0.368	133,581
Politically Connected Director	0.197	0	0.397	133,581
Panel B: Firm Characteristics				
	Mean	Median	Std. Dev.	No. of Obs.
MTB	2.377	1.94	1.423	13,722
TFP	0	-0.008	0.265	12,734

Table III, part 1

1642

The Journal of Finance[®]

Table III—Continued

Panel B: Firm Characteristics				
	0.046	0.065	0.191	13,294
ROE	0.081	0	0.111	13,840
Foreign Experience	0.77	0	1.069	13,840
No. of Directors with Foreign Experience	0.463	0	0.499	13,840
Directors with Foreign Experience Dummy	9.448	9	2.111	13,840
Board Size	0.703	1	0.457	13,840
State	0.023	0	0.073	13,840
Foreign Ownership	0.403	0.386	0.166	13,840
Block	32.636	14.781	59.287	13,840
Assets (RMB 100 millions)	0.5	0.489	0.235	13,840
Leverage	1.982	2	0.833	13,840
Director Tenure	48.113	48.143	3.977	13,840
Director Age	16.0%	12.5%	14.3%	13,840
Busy Directors	39.7%	37.5%	22.1%	13,840
Non-Independent Directors	9.7%	9.1%	10.3%	13,840
Female Directors	0.2%	0.0%	1.6%	13,840
Foreign Directors	19.5%	16.7%	16.5%	13,840
Board Political Connection	4.719	5	0.441	13,840
Average Chinese University Rank	2.199	2	1.406	13,840
No. of Business Segments	-0.014	0.006	0.093	13,840
Free Cash Flow	0.269	0	0.444	13,840
Young IPO Firm	0.034	0.029	0.021	13,840
Stock Volatility	45.822	45	6.651	13,644
CEO Tenure	3.265	3	2.42	13,743
CEO Age	0.147	0	0.354	13,826
CEO Turnover	0.057	0	0.233	4,094
Foreign M&A	0.132	0	0.339	357
Foreign Private Placement	0.121	0	0.214	5,917
Foreign Sales	-0.001	-0.001	0.091	13,084
Earnings Management	Panel C: Industry Distribution of Sample Firms			
Industry			%	No. of Obs.
Agriculture			2.48%	343
Mining			1.64%	227
Food			4.32%	598
Apparel			4.51%	624
Furniture			0.25%	34
Printing			2.02%	280
Gas and chemistry			11.12%	1,539
Electronic			3.63%	503
Metal			9.39%	1,300
Machinery			15.61%	2,161
Pharmaceutical products			6.29%	871
Other manufacturing			1.46%	202

(Continued)

Table III, part 2

Table III—Continued

Panel C: Industry Distribution of Sample Firms		
Energy supply	4.15%	575
Construction	1.94%	269
Transportation	4.12%	570
Information technology	5.77%	798
Retail and wholesale	7.16%	991
Real estate	4.09%	566
Other service supply	3.16%	438
Entertainment	0.86%	119
Other	6.01%	832

factor productivity (TFP).¹³ As is common in the literature (see, for instance, Schoar (2002)), we compute a firm's total factor productivity as the residual, $\hat{\varepsilon}_{it}$ of the firm level regression

$$y_{ijt} = \alpha_{jt} + \beta_{jt}l_{ijt} + \gamma_{jt}k_{ijt} + \delta_{jt}m_{ijt} + \varepsilon_{ijt}, \quad (1)$$

where y_{ijt} is the logarithm of the sales of firm i belonging to industry j during year t , l_{ijt} is the logarithm of the number of workers of firm i during year t , k_{ijt} is the logarithm of the total assets of firm i during year t , and m_{ijt} is the logarithm of the expenses for material and other inputs of firm i during year t . We estimate all equations by industry and year. For this reason, our estimate of total factor productivity captures a firm's deviation from the factor productivity within its industry in a given year. We also report the return on assets (ROA), board size, leverage, and state ownership, which are broadly consistent with existing studies (see Cao et al. (2011), Jiang, Wan, and Zhao (2013)).

7.4 Table IV: Policy Changes and the Board of Directors

Read:

- The dependent variable is the proportion of directors with foreign experience.
- Provincial policy adoption predicts a higher proportion of directors with foreign experience.
- Ownership interactions show that some ownership structures may be better positioned to use the new supply of returnee talent.
- The table reports first-stage strength statistics for the policy instruments.

Detailed interpretation: This is the first-stage table. Its role is to show that the staggered policy adoption affects actual board composition, not merely the general labor market. The estimates support the use of the policies as instruments in later performance regressions.

Economic message: Provincial talent policy changes the composition of corporate boards, making brain-gain effects measurable at the firm level.

Table IV
Policy Changes and the Board of Directors

This table relates the proportion of directors with foreign experience ("Foreign Experience") to firm characteristics and the provincial policies. "Provincial Policy" is a dummy that takes a value of one if the firm is headquartered in a given province in the years following the adoption of a policy to encourage the return of highly skilled emigrants, and zero otherwise. MTB ($t-1$), ROE ($t-1$), MTB ($t-2$), and ROE ($t-2$) are MTB and ROE lagged for one and two years, respectively. All other variables are defined in the Appendix. t -statistics, computed with robust standard errors clustered at the year level, are reported in parentheses. All models include a constant, different sets of fixed effects, and province-specific linear trends, as indicated in the table, but the coefficients are not reported. We also report an F -test to assess the joint significance of the policy dummies (F -test of excluded instruments). ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Foreign Ownership	0.474*** (25.76)	0.473*** (24.55)	0.456*** (25.77)	0.461*** (25.03)	0.474*** (25.67)	0.413*** (8.68)	0.473*** (24.43)	0.413*** (8.56)
Block	-0.014** (-2.51)	-0.016** (-2.38)	-0.018* (-1.82)	-0.019* (-1.88)	-0.014** (-2.43)	-0.012* (-2.20)	-0.016** (-2.39)	-0.015** (-2.46)
State	-0.012*** (-3.94)	-0.012*** (-3.79)	-0.011** (-3.12)	-0.011** (-2.89)	-0.012*** (-3.93)	-0.013*** (-3.40)	-0.012*** (-3.76)	-0.014*** (-3.48)
Size	0.008*** (6.59)	0.010*** (6.41)	0.007*** (7.62)	0.008*** (3.94)	0.008*** (6.54)	0.008*** (6.54)	0.008*** (6.40)	0.008*** (6.32)
Leverage	-0.001 (-0.26)	0.004 (1.57)	0.004 (1.57)	0.004* (2.19)	0.004* (2.19)	-0.001 (-0.30)	-0.001 (-0.26)	-0.001 (-0.26)
# of Business Segments	-0.000 (-0.17)	-0.000 (-0.69)	-0.000 (-0.69)	-0.000 (-0.32)	-0.000 (-0.32)	-0.000 (-0.25)	-0.000 (-0.25)	-0.000 (-0.25)
Free Cash Flow	0.003 (0.24)	0.003 (0.75)	0.003 (0.75)	0.015 (1.30)	0.015 (1.30)	0.004 (0.52)	0.004 (0.53)	0.004 (0.53)
Young IPO Firm	(1.17)	(0.33)	(0.01)	(0.26)	(0.26)	(1.35)	(1.55)	(1.55)
Stock Volatility	-0.107 (-1.17)	-0.503 (-1.70)	-0.503 (-1.70)	-0.567 (-1.58)	-0.567 (-1.58)	-0.105 (-1.76)	-0.105 (-1.73)	-0.105 (-1.73)
MTB ($t-1$)								

(Continued)

The Brain Gain of Corporate Boards

Table IV—Continued

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
MTB ($t-2$)				0.002 (0.75)				
ROE ($t-1$)					0.007 (0.80)			
ROE ($t-2$)						0.002 (0.29)		
Provincial Policy					0.011** (2.99)	0.009** (2.27)	0.012** (3.11)	0.007* (1.96)
Provincial Policy × Foreign Ownership						0.068* (1.59)		0.092* (1.58)
Provincial Policy × Block						-0.002 (-0.29)		-0.000 (-0.06)
Provincial Policy × State						0.003 (0.58)		0.003 (0.92)
Province-Specific Trend	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F -test of excluded instruments					8.96**	6.75***	9.67***	6.60***
No. of obs.	13,840	13,840	12,053	11,891	13,840	13,840	13,840	13,840
R^2	0.188	0.189	0.183	0.185	0.189	0.191	0.190	0.191

1646 The Journal of Finance®

Table IV, part 1

Table IV, part 2

7.5 Table V: Directors with Foreign Experience and Firm Value

Read:

- The dependent variable is industry-year adjusted MTB.
- OLS and IV estimates both point to a positive effect of directors with foreign experience on firm valuation.
- The IV estimates are large, consistent with a strong effect for firms whose boards respond to local supply shocks.
- Robustness checks include within-province tests, controls for engaged firms, and exclusions of firms that never hire such directors.

Detailed interpretation: Table V is the headline valuation result. It shows that the market values firms more highly when foreign-experienced directors enter the board, and that this is not solely an OLS selection pattern.

Economic message: Investors appear to price foreign-experienced directors as valuable governance and advisory capital.

Table V Directors with Foreign Experience and Firm Value								
This table relates the firm's MTB to the proportion of directors with foreign experience. In columns 1 to 3, we present ordinary least squares estimates. In columns 4 to 8, we present instrumental variable estimates. The instrumental variables include "Provincial Policy," a dummy that takes a value of one in years following the implementation of the policy in each province, and interaction variables between the policy dummy and firm ownership characteristics "State," "Foreign Ownership," and "Block" in 1999 (the beginning of the sample period). If a firm enters our sample later than 1999, firm ownership characteristics computed in the year of the firm's entry in the sample are used to construct the interaction terms. In columns 1 to 5 and 7 to 8, the dependent variable is the firm's MTB, from which we subtract the industry-year median. The dependent variable in column 6 is the firm's MTB, from which we subtract the industry-year median and the province-year median. In column 8, we exclude all firms that do not hire directors with foreign experience during the sample period. All the variables are defined in the Appendix. <i>t</i> -statistics computed with robust standard errors clustered at the firm level are reported in parentheses. All models include a constant, but the coefficients are not reported. We report the partial <i>R</i> ² of the instruments in the first stage and the Craig-Donald Wald <i>F</i> -statistics. We also report the Stock-Yogo weak ID test critical values for the Craig-Donald Wald <i>F</i> -statistics. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.								
	OLS			IV				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
						Within-Province Test	Including Firm FE and Lagged Performance	Excluding Firms that Never Hired Directors with Foreign Experience
Foreign Experience	0.900*** (5.66)	0.805*** (5.56)	0.480*** (2.80)	4.092*** (3.13)	7.310*** (4.17)	17.505*** (5.56)	11.350*** (4.84)	8.250*** (4.61)
Foreign Ownership		0.376 (1.48)	0.710 (1.45)	-1.278* (-1.92)	-2.835*** (-3.29)	-8.047*** (-4.68)	-5.617*** (-3.44)	-3.970*** (-3.16)
Block		0.540*** (5.40)	-0.228 (-1.32)	0.650*** (5.71)	0.670*** (4.89)	1.147*** (4.41)	0.323 (1.19)	0.341 (1.21)
State		-0.127*** (-3.48)	-0.078 (-1.56)	-0.065 (-1.35)	-0.030 (-0.50)	0.065 (0.60)	0.112 (1.38)	0.050 (0.65)
Size	-0.507*** (-24.99)	-0.524*** (-24.67)	-0.679*** (-17.96)	-0.560*** (-19.66)	-0.583*** (-17.47)	-0.659*** (-10.93)	-0.732*** (-14.20)	-0.743*** (-13.98)
Leverage	0.071 (0.67)	0.201** (2.05)	0.597*** (5.11)	0.243** (2.04)	0.235* (1.82)	0.293 (1.53)	0.577*** (4.52)	0.546*** (4.12)

(Continued)

Table V—Continued								
	OLS			IV				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
No. of Business Segments	-0.028*** (-2.81)	-0.008 (-0.68)	-0.034*** (-3.02)	-0.033** (-2.51)	-0.019 (-0.76)	-0.005 (-0.30)	-0.002 (-0.12)	-0.002 (-0.12)
Free Cash Flow	0.970*** (6.00)	0.800*** (5.71)	0.950*** (5.50)	0.880*** (5.50)	-0.177 (-0.61)	0.833*** (4.60)	0.788*** (4.05)	0.788*** (4.05)
Young IPO Firm	-0.142*** (-4.77)	-0.163*** (-4.86)	-0.137*** (-4.36)	-0.113*** (-3.08)	-0.306*** (-4.69)	-0.070 (-1.59)	-0.071 (-1.50)	-0.071 (-1.50)
Stock Volatility	5.774*** (11.01)	5.359*** (11.70)	5.762*** (10.63)	5.804*** (9.81)	2.445*** (2.71)	13.320*** (7.29)	12.872*** (7.02)	12.872*** (7.02)
MTB (<i>t</i> - 1)						0.176*** (12.54)	0.174*** (11.87)	0.174*** (11.87)
Firm FE	No	No	Yes	No	No	Yes	Yes	Yes
Province FE	No	No	No	No	No	No	No	No
Partial <i>R</i> ²				0.011	0.008	0.011	0.009	0.014
Craig-Donald Wald <i>F</i> -statistic				39.363	27.331	39.363	26.485	30.084
5% maximal IV				16.85	16.85	16.85	16.85	16.85
10% maximal IV				10.27	10.27	10.27	10.27	10.27
relative bias	13.722	13.722	13.722	13.722	13.722	13.722	12.862	9.798
No. of obs.	0.194	0.218	0.125	0.214	0.233	0.159	0.172	0.175
<i>R</i> ²								

The Brain Gain of Corporate Boards

1651

Table V, part 1

Table V, part 2

7.6 Table VI: Directors with Foreign Experience and Total Factor Productivity

Read:

- The dependent variable is industry-year adjusted TFP.
- Foreign-experienced directors predict higher productivity in both OLS and IV specifications.
- Robustness specifications parallel the valuation tests.

Detailed interpretation: This table links the board shock to real operating performance. The result is important because valuation could reflect expectations or investor sentiment, while TFP points to improvements in firm operations.

Economic message: Directors with foreign experience seem to affect how firms operate, not only how markets perceive them.

Table VI
Directors with Foreign Experience and Total Factor Productivity

This table relates the firm's TFP to the presence of directors with foreign experience. In columns 1 to 3, we present ordinary least squares estimates. In columns 4 to 8, we present instrumental variable estimates. The instrumental variables in columns 4 to 8 include "Provincial Policy," a dummy that takes a value of one in years following the implementation of the policy in each province, and interaction variables between the policy dummy and firm ownership characteristics "State," "Foreign Ownership," and "Block" in 1999 (the beginning of the sample period). If a firm enters our sample later than 1999, firm ownership characteristics computed in the year of the firm's entry in the sample are used to construct the interaction terms. In columns 1 to 5, and 7 to 8, the dependent variable is the firm's TFP, from which we subtract the industry-year median. The dependent variable in column 6 is the firm's TFP, from which we subtract the industry-year median and the province-year median. In column 8, we exclude all firms that do not hire directors with foreign experience during the sample period. All the variables are defined in the Appendix. *t*-statistics computed with robust standard errors clustered at the firm level are reported in parentheses. All models include a constant, but the coefficients are not reported. We report the partial R^2 of the instruments in the first stage and the Crag-Donald Wald F -statistics. We also report the Stock-Yogo weak ID test critical values for the Crag-Donald Wald F -statistics. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	OLS			IV			
	(1)	(2)	(3)	(4)	(5)	(6)	(8)
				Within Province Test	Including Firm FE and Lagged Performance	Excluding Firms that Never Hired Directors with Foreign Experience	
Foreign Experience	0.111*** (2.78)	0.134*** (3.10)	0.089** (2.30)	0.975*** (2.90)	0.822** (2.33)	0.797** (2.43)	0.903** (2.33)
Foreign Ownership		-0.050 (-0.79)	-0.088 (-0.86)	-0.490*** (-2.66)	-0.431** (-2.44)	-0.517* (-1.73)	-0.485** (-1.97)
Block		0.063** (2.10)	0.037 (0.96)	0.099*** (2.81)	0.076** (2.37)	0.074** (2.32)	0.080 (1.59)
State		0.009 (0.30)	0.005 (0.42)	0.028** (2.14)	0.024* (1.94)	0.025* (1.95)	0.035** (2.06)
Size	0.001 (0.31)	-0.005 (-1.22)	-0.041*** (-5.59)	-0.014** (-2.30)	-0.013** (-2.11)	-0.051*** (-3.84)	-0.057*** (-3.86)
Leverage		-0.134*** (-6.74)	-0.003 (-1.40)	-0.029 (-1.28)	-0.003 (-1.49)	0.002 (0.07)	0.008 (0.92)

(Continued)

Table VI, part 1

Table VI—Continued

	OLS			IV			
	(1)	(2)	(3)	(4)	(5)	(6)	(8)
						Within Province Test	Including Firm FE and Lagged Performance
No. of Business Segments	-0.012*** (-4.28)	-0.004 (-1.38)	-0.014*** (-4.45)	-0.013*** (-4.25)	-0.015*** (-4.75)	-0.005* (-1.67)	-0.007** (-2.23)
Free Cash Flow	0.574*** (15.48)	0.417*** (12.26)	0.556*** (13.99)	0.549*** (14.33)	0.532*** (13.78)	0.382*** (9.92)	0.386*** (9.07)
Young IPO Firm	0.016** (1.99)	-0.004 (-0.56)	0.016* (1.89)	0.016* (1.87)	0.014* (1.69)	0.001 (0.16)	-0.003 (-0.32)
Stock Volatility	-0.048 (-0.47)	0.087 (0.84)	-0.051 (-0.47)	-0.053 (-0.50)	0.056 (0.54)	0.052*** (2.61)	0.024*** (2.28)
TFP ($t-1$)						0.208*** (11.96)	0.208*** (11.19)
Firm FE	No	No	Yes	No	No	Yes	Yes
Province FE	No	No	No	No	No	No	No
Partial R^2			0.011	0.008	0.011	0.006	0.009
Crag-Donald Wald			34.354	24.580	34.354	14.417	16.731
F -statistic							
5% maximal IV relative bias			16.85	16.85	16.85	16.85	16.85
10% maximal IV relative bias			10.27	10.27	10.27	10.27	10.27
No. of obs.	12,734	12,734	12,734	12,734	12,734	11,074	8,429
R^2	0.015	0.054	0.036	0.053	0.064	0.050	0.083

Table VI, part 2

7.7 Table VII: Directors with Foreign Experience and Profitability

Read:

- The dependent variable is future industry-year adjusted ROE.
- Effects are positive in several specifications, including IV models.
- Profitability responds after the board-composition change, consistent with gradual implementation of improved practices or strategies.

Detailed interpretation: Table VII complements the valuation and productivity tables. It shows that the change in board human capital ultimately translates into accounting profitability.

Economic message: A returnee director is not merely a symbolic signal; the firm appears to become more profitable after the board changes.

Table VII
Directors with Foreign Experience and Profitability

This table relates the firm's profitability to the presence of directors with foreign experience. In columns 1 to 3, we present ordinary least squares estimates. In columns 4 to 8, we present instrumental variable estimates. The instrumental variables in columns 4 to 8 include "Provincial Policy," a dummy that takes a value of one in years following the implementation of the policy in each province, and interaction variables between the policy dummy and firm ownership characteristics "State," "Foreign Ownership," and "Block" in 1999 (the beginning of the sample period). If a firm enters our sample later than 1999, firm ownership characteristics computed in the year of the firm's entry in the sample are used to construct the interaction terms. In columns 1 to 5, and 7 to 8, the dependent variable is the firm's ROE at $t+1$, from which we subtract the industry-year median. The dependent variable in column 6 is the firm's ROE at $t+1$, from which we subtract the industry-year median and the province-year median. In column 8, we exclude all firms that do not hire directors with foreign experience during the sample period. All the variables are defined in the Appendix. *t*-statistics computed with robust standard errors clustered at the firm level are reported in parentheses. All models include a constant, but the coefficients are not reported. We report the partial R^2 of the instruments in the first stage and the Crag-Donald Wald F -statistics. We also report the Stock-Yogo weak ID test critical values for the Crag-Donald Wald F -statistics. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	OLS			IV			
	(1)	(2)	(3)	(4)	(5)	(6)	(8)
				Within Province Test	Including Firm FE and Lagged Performance	Excluding Firms that Never Hired Directors with Foreign Experience	
Foreign Experience	0.053*** (2.81)	0.057*** (3.03)	0.031 (1.06)	0.541*** (2.82)	0.556** (2.54)	0.362** (1.99)	0.430* (1.85)
Foreign Ownership		-0.040 (-1.27)	0.039 (0.47)	-0.284** (-2.57)	-0.289** (-2.46)	-0.209** (-2.00)	-0.192 (-1.57)
Block		0.050*** (4.05)	0.122*** (4.69)	0.066*** (4.30)	0.068*** (4.31)	0.066*** (4.56)	0.148*** (4.36)
State		-0.018*** (-3.67)	-0.031*** (-3.43)	-0.009 (-1.45)	-0.006 (-0.97)	-0.009 (-1.55)	-0.015 (-1.25)
Size	0.019*** (8.66)	0.018*** (8.05)	-0.039*** (-7.41)	0.013*** (3.80)	0.013*** (3.67)	0.012*** (3.97)	-0.034*** (-5.58)

(Continued)

Table VII, part 1

Table VII—Continued

	OLS			IV			
	(1)	(2)	(3)	(4)	(5)	(6)	(8)
						Within Province Test	Including Firm FE and Lagged Performance
Leverage	-0.055*** (-3.69)	0.001 (0.09)	0.212*** (7.63)	0.005 (0.30)	0.003 (0.17)	-0.000 (-0.03)	0.113*** (4.64)
No. of Business Segments		0.002 (1.31)	-0.000 (-0.11)	0.001 (0.52)	0.001 (0.61)	0.001 (0.84)	-0.001 (-0.43)
Free Cash Flow	0.363*** (12.94)	0.232*** (7.58)	0.345*** (12.14)	0.347*** (12.11)	0.323*** (11.50)	0.323*** (4.78)	0.175*** (4.81)
Young IPO Firm	0.032*** (8.34)	0.011** (2.09)	0.033*** (7.80)	0.030*** (7.23)	0.031*** (7.85)	0.014** (2.33)	0.010 (1.52)
Stock Volatility	0.103** (1.98)	0.250*** (3.90)	0.067* (1.74)	0.063* (1.66)	0.055 (0.99)	0.055 (0.99)	0.227*** (3.95)
ROE						0.094*** (4.11)	0.084*** (3.29)
Firm FE	No	No	Yes	No	No	No	Yes
Province FE	No	No	No	No	Yes	No	No
Partial R^2			0.013	0.009	0.013	0.009	0.013
Crag-Donald Wald			42.664	29.009	42.664	24.602	28.505
F -statistic							
5% maximal IV relative bias			16.85	16.85	16.85	16.85	16.85
10% maximal IV relative bias			10.27	10.27	10.27	10.27	10.27
No. of obs.	13,294	13,294	13,294	13,294	13,294	13,139	9,972
R^2	0.013	0.045	0.030	0.045	0.051	0.039	0.026

Table VII, part 2

7.8 Table VIII: Powerful Firms and Directors with Foreign Experience

Read:

- The table examines whether political or economic power drives the results.
- Interactions test whether the foreign-experience effect differs for powerful versus less powerful firms.

- The analysis helps address the concern that only politically connected or economically powerful firms benefit from provincial policies.

Detailed interpretation: The table asks whether board brain gain is just another form of elite access. The evidence suggests that foreign-experienced directors remain relevant after accounting for political and economic power.

Economic message: The value of international human capital is not reducible to local political influence.

Table VIII
Powerful Firms and Directors with Foreign Experience

In columns 1 to 3, we capture a firm's political influence using the presence of politically connected directors. The dummy "No Power" takes the value of one if a firm's board does not have politically connected directors. In columns 4 to 6, we measure a firm's economic power using the number of employees. The dummy "No Power" takes the value of one if the number of employees is less than 2,000. The State-Owned Assets Supervision and Administration Commission of the State Council of China considers large firms those with more than 2,000 employees. The dependent variables are the firm's MTB, TFP, and ROE at $t + 1$, from which we subtract the industry-year median. The instruments for "Foreign Experience" and "Foreign Experience" $\times$ "No Power" are "Provincial Policy," a dummy that takes a value of one in years following the implementation of the policy in each province, interaction variables between the policy dummy and firm ownership characteristics "State," "Foreign Ownership," and "Block" in 1999 (the beginning of the sample period), and interactions of all the instruments mentioned above with the dummy "No Power." If a firm enters our sample later than 1999, "State," "Foreign Ownership," and "Block" computed in the year of the firm's entry in the sample are used to construct the interaction terms. All the variables are defined in the Appendix. t -statistics computed with robust standard errors clustered at the firm level are reported in parentheses. All models include a constant, but the coefficients are not reported. We report the partial R^2 of the instruments in the first stage and the Cragg-Donald Wald F -statistics. We also report the Stock-Yogo weak ID test critical values for the Cragg-Donald Wald F -statistics. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	Political Power			Economic Power		
	MTB	TFP	ROE ($t + 1$)	MTB	TFP	ROE ($t + 1$)
	(1)	(2)	(3)	(4)	(5)	(6)
Foreign Experience	4.453*** (3.27)	1.066*** (2.99)	0.476** (2.42)	3.722*** (3.36)	0.846*** (2.74)	0.596*** (3.38)
Foreign Experience $\times$ No Power	-1.117 (-1.02)	-0.292 (-1.44)	0.082 (0.72)	-1.120 (-1.08)	0.290 (1.34)	0.085 (0.73)
No Power	0.015 (0.17)	0.022 (1.16)	-0.003 (-0.30)	0.201** (2.12)	-0.044** (-2.01)	-0.017 (-1.50)
Foreign Ownership	-1.285* (-1.90)	-0.490*** (-2.68)	-0.264** (-2.48)	-0.821 (-1.25)	-0.491*** (-2.78)	-0.351*** (-3.15)
Block	0.657*** (5.65)	0.091*** (2.72)	0.064*** (4.27)	0.666*** (6.14)	0.054*** (2.62)	0.064*** (4.14)
State	-0.069 (-1.43)	0.028** (2.18)	-0.010 (-1.56)	-0.074 (-1.62)	0.027** (2.18)	-0.008 (-1.21)

(Continued)

Table VIII, part 1

Table VIII—Continued

	Political Power			Economic Power		
	MTB	TFP	ROE ($t + 1$)	MTB	TFP	ROE ($t + 1$)
	(1)	(2)	(3)	(4)	(5)	(6)
Size	-0.254*** (-19.79)	-0.015** (-2.36)	0.013*** (4.11)	-0.529*** (-17.67)	-0.017** (-2.51)	0.010*** (2.58)
Leverage	0.241** (2.03)	-0.029 (-1.28)	0.004 (0.28)	0.248** (2.10)	-0.030 (-1.31)	0.005 (0.33)
No. of Business Segments	-0.035*** (-3.12)	-0.014*** (-4.44)	0.001 (0.61)	-0.034*** (-3.15)	-0.014*** (-4.44)	0.000 (0.27)
Free Cash Flow	0.933*** (5.42)	0.553*** (13.80)	0.350*** (12.24)	0.918*** (5.47)	0.559*** (13.96)	0.347*** (11.83)
Young IPO Firm	-0.138*** (-4.34)	0.016* (1.85)	0.033*** (7.96)	-0.138*** (-4.47)	0.017* (1.94)	0.033*** (7.58)
Stock Volatility	5.793*** (10.66)	-0.054 (-3.15)	0.096* (1.75)	5.672*** (10.49)	-0.033 (-3.20)	0.105* (1.78)
Partial R^2	0.013	0.012	0.014	0.013	0.013	0.015
Cragg-Donald Wald F -statistic	19.684	17.134	21.133	21.336	19.112	23.068
5% maximal IV relative bias	17.70	17.70	17.70	17.70	17.70	17.70
10% maximal IV relative bias	10.22	10.22	10.22	10.22	10.22	10.22
No. of obs.	13,722	12,734	13,294	13,594	12,734	13,168
R^2	0.215	0.053	0.045	0.217	0.053	0.046

Table VIII, part 2

7.9 Table IX: Proportion of Directors with Foreign Experience and Board Characteristics

Read:

- The table adds extensive board controls, including director tenure, employed directors, female directors, busy directors, foreign directors, political connections, board size, and Chinese university ranking.
- Foreign experience continues to predict MTB, TFP, and ROE.
- Foreign directors without a Chinese returnee background do not appear to have the same positive effect.

Detailed interpretation: Table IX addresses a central alternative explanation: perhaps foreign-experienced directors are just better-educated or are part of broader board upgrades. The results remain after controlling for board structure and ability proxies.

Economic message: The paper's effect is specifically about foreign-experienced human capital on the board, not merely about board size, independence, or generic talent.

Table IX
Proportion of Directors with Foreign Experience and Board Characteristics

We present the instrumental variable estimates relating firm performance to the presence of directors with foreign experience controlling for additional board characteristics. In columns 1 to 3, the dependent variables are the firm's MTB, TFP, and ROE at $t + 1$, from which we subtract the industry-year median. The instruments for "Foreign Experience" are "Provincial Policy," a dummy that takes a value of one in years following the implementation of the policy in each province, and interaction variables between the policy dummy and firm ownership characteristics "State," "Foreign Ownership," and "Block" in 1999 (the beginning of the sample period). If a firm enters our sample later than 1999, "State," "Foreign Ownership," and "Block" computed in the year of the firm's entry in the sample are used to construct the interaction terms. Additional board characteristics include "Average Director Tenure," "Employed Directors," "Female Directors," "Busy Directors," "Foreign Directors," "Average Director Age," "Board Size," "Board Political Connection," and "Average Rank of Chinese Universities." All the variables are defined in the Appendix. t -statistics computed with robust standard errors clustered at the firm level are reported in parentheses. All models include a constant, but the coefficients are not reported. We report the partial R^2 of the instruments in the first stage and the Cragg-Donald Wald F -statistics. We also report the Stock-Yogo weak ID test critical values for the Cragg-Donald Wald F -statistics. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	MTB	TFP	ROE ($t + 1$)
	(1)	(2)	(3)
Foreign Experience	4.311** (2.00)	1.040** (2.01)	0.639** (2.10)
Foreign Ownership	-0.894 (-1.16)	-0.408** (-2.04)	-0.258** (-2.05)
Block	0.639*** (5.67)	0.083*** (2.58)	0.062*** (4.10)
State	-0.109* (-1.96)	0.019 (1.32)	-0.009 (-1.24)
Size	-0.565*** (-20.01)	-0.019*** (-3.08)	0.013*** (4.00)
Leverage	0.240** (2.02)	-0.026 (-1.16)	0.006 (0.35)
No. of Business Segments	-0.037*** (-3.13)	-0.014*** (-4.30)	0.000 (0.09)
Free Cash Flow	0.931*** (5.49)	0.554*** (14.03)	0.346*** (11.93)
Young IPO Firm	-0.132*** (-4.12)	0.014 (1.57)	0.033*** (7.54)
Stock Volatility	5.804*** (10.51)	-0.043 (-0.38)	0.098 (1.58)
Average Director Tenure	0.021 (1.59)	0.002 (0.48)	0.004** (2.10)
Non Independent Directors	-0.204* (-1.80)	0.049* (1.73)	0.020 (1.22)
Board Size	0.013* (1.73)	0.006*** (2.81)	-0.000 (-0.19)
Female Directors	-0.176 (-0.98)	-0.016 (-0.37)	0.040 (1.57)
Average Director Age	0.004	0.004***	0.001

(Continued)

Table IX, part 1

Table IX—Continued

	MTB	TFP	ROE ($t + 1$)
	(1)	(2)	(3)
Foreign Directors	(0.85) -5.306* (-1.67)	(2.88) -1.241 (-1.64)	(0.90) -0.648 (-1.53)
Board Political Connection	0.211* (1.96)	-0.011 (-0.34)	0.010 (0.64)
Average Rank of Chinese Universities	0.150 (1.32)	0.050* (1.81)	0.035** (2.21)
Busy Directors	-0.143 (-0.60)	0.004 (0.08)	0.001 (0.02)
Partial R^2	0.006	0.005	0.006
Cragg-Donald Wald F -statistic	18.808	16.751	20.979
5% maximal IV relative bias	16.85	16.85	16.85
10% maximal IV relative bias	10.27	10.27	10.27
No. of obs.	13,722	12,734	13,294
R^2	0.224	0.060	0.048

foreign directors cannot be effective monitors, even in a high transparency environment like the United States, and hence are extremely rare.

In an alternative test, we first conjecture that foreign experience may be more important for directors specializing in business and management because the gap in academic standards between China and the rest of the world has generally been smaller in scientific subjects, such as engineering or physics. We then ask whether firms with at least one director with a foreign business degree perform better than firms that have directors with foreign experience but no director with a foreign business degree.

In Table X, the coefficient on the dummy capturing the presence of directors with foreign business degrees is always larger than that for the analogous dummy capturing other foreign degrees (even though the difference is not statistically significant), suggesting that foreign experience matters. However, in interpreting this finding, and the other tests below exploiting heterogeneity of directors' foreign experience, it must be kept in mind that our instruments can explain changes in the proportion of directors with foreign experience but not in the kind of foreign experience acquired by the directors. Thus, we can only provide suggestive evidence based on ordinary least squares estimates.

V. What Do Directors with Foreign Experience Do?

The causal mechanism behind our maintained hypothesis that directors with foreign experience positively affect firm performance implies that the way in which firms are run changes when directors with foreign experience join the board. To provide evidence on the mechanisms driving our results, we explore

Table IX, part 2

7.10 Table X: Business Education and Other Foreign Experience

Read:

- The table separates economics/business foreign degrees from other types of foreign experience.
- It asks whether the gains are concentrated among directors whose foreign experience is most directly relevant to business decisions.
- The results help distinguish general cosmopolitan exposure from task-specific expertise.

Detailed interpretation: Business-related foreign education and experience are especially relevant if directors improve corporate governance, management practices, and international business decisions. The table therefore strengthens the mechanism interpretation.

Economic message: The kind of foreign experience matters; business-relevant experience is particularly valuable for corporate boards.

7.11 Table XI: Internationalization

Read:

Table X
Business Education and Other Foreign Experience

This table presents ordinary least squares estimates relating firm performance to directors' foreign experience. "Foreign Economics/Business Degree" is a dummy equal to one if at least one director earned a foreign degree with an economics or business major, and zero otherwise. "Foreign Non-Economics/Business Degree" is a dummy equal to one if none of the directors with foreign experience have a foreign degree with an economics or business major. The dependent variables are the firm's MTB, TFP, and ROE at $t + 1$, from which we subtract the industry-year median. All the variables are defined in the Appendix. t -statistics computed with robust standard errors clustered at the firm level are reported in parentheses. All models include a constant, but the coefficients are not reported. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	MTB	TFP	ROE($t + 1$)
	(1)	(2)	(3)
Foreign Economics/Business Degree	0.106*** (3.38)	0.026*** (2.80)	0.012*** (2.82)
Foreign Non-Economics/Business Degree	0.091** (2.29)	0.018* (1.75)	0.009* (1.79)
Foreign Ownership	0.621** (2.55)	-0.015 (-0.26)	-0.028 (-0.91)
Block	0.539*** (5.41)	0.062** (2.09)	0.050*** (4.05)
State	-0.136*** (-3.70)	0.008 (0.85)	-0.018*** (-3.94)
Size	-0.525*** (-24.56)	-0.006 (-1.38)	0.018*** (7.82)
Leverage	0.236** (2.10)	-0.028 (-1.33)	0.002 (0.12)
No. of Business Segments	-0.027*** (-2.75)	-0.012*** (-4.28)	0.002 (1.33)
Free Cash Flow	0.987*** (6.03)	0.574*** (15.52)	0.364*** (12.94)
Young IPO Firm	-0.141*** (-4.74)	0.016** (2.00)	0.032*** (8.38)
Stock Volatility	5.768*** (10.98)	-0.050 (-0.48)	0.103** (1.97)
No. of obs.	13,722	12,734	13,294
R^2	0.215	0.054	0.045

- The table studies foreign M&A, foreign private placements, and foreign sales.
- Firms with foreign-experienced directors are more likely to pursue international transactions.
- Same-country interactions show that international activity is especially likely in countries where the directors obtained experience.

Detailed interpretation: This is the connections-and-advice channel. The same-country evidence is particularly informative because it links the geography of director experience to the geography of firm internationalization.

Economic message: Foreign-experienced directors help firms access foreign investment, financing, and product markets.

Table XI
Internationalization

The dependent variable in columns 1 to 3 is "Foreign M&A," a dummy equal to one if at least one of the mergers and acquisitions that a sample firm announced in a given year involves a foreign firm. In columns 4 and 5, the dependent variable is "Foreign Private Placement," a dummy equal to one if at least one of the firm's private placements in a given year is targeted at foreign investors, and zero if none of these private placements is targeted at foreign investors. In columns 6 to 8, the dependent variable is "Foreign Sales," from which we subtract the industry-year median. "Same Country" is a dummy equal to one if at least one director obtained higher foreign experience in the same country as the country of the selling firm, the foreign investors, and the foreign sales, respectively. The instrumental variables in columns 2 and 7 include "Provincial Policy," a dummy that takes a value of one in years following the implementation of the policy in each province, and interaction variables between the policy dummy and firm ownership characteristics "State," "Foreign Ownership," and "Block" in 1999. If a firm enters our sample later than 1999, firm ownership characteristics computed in the year of the firm's entry in the sample are used to construct the interaction terms. Variable definitions are in the Appendix. t -statistics computed with robust standard errors clustered at the firm level are reported in parentheses. All models include a constant, but the coefficients are not reported. We report the partial R^2 of the instruments in the first stage and the Cragg-Donald Wald F -statistics. We also report the Stock-Yogo weak ID test critical values for the Cragg-Donald Wald F -statistics. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	Foreign M&A			Foreign Private Placement		Foreign Sales		
	OLS	IV	OLS	OLS	OLS	OLS	IV	OLS
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Foreign Experience	0.154*** (3.22)	0.899* (1.76)	0.007 (0.18)	0.319* (1.66)	0.076 (0.42)	0.216*** (3.35)	0.633** (2.14)	0.176*** (2.72)
Foreign Experience × Same Country			2.900*** (14.26)		2.969*** (5.83)			1.193*** (3.82)
Foreign Ownership	0.060*** (4.51)	0.005 (0.12)	0.042*** (3.54)	0.006 (0.12)	-0.004 (-0.09)	0.043*** (2.70)	0.014 (0.62)	0.038*** (2.41)
Block	-0.054** (-2.05)	-0.040 (-1.32)	-0.036 (-1.59)	-0.118 (-1.01)	-0.096 (-0.89)	0.021 (0.61)	0.039 (1.07)	0.021 (0.62)
State	-0.012 (-1.34)	-0.000 (-0.03)	-0.007 (-0.80)	-0.052 (-1.11)	-0.007 (-0.16)	0.069 (0.78)	0.016 (1.21)	0.010 (0.87)
Size	0.009* (1.92)	0.000 (0.05)	0.006 (1.54)	0.026 (1.49)	0.028* (1.71)	-0.007 (-1.30)	-0.010* (-1.70)	-0.007 (-1.57)

(Continued)

Table XI—Continued

	Foreign M&A			Foreign Private Placement		Foreign Sales		
	OLS	IV	OLS	OLS	OLS	OLS	IV	OLS
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Leverage	-0.011 (-0.52)	-0.020 (-0.80)	-0.005 (-0.27)	-0.147 (-1.53)	-0.133 (-1.48)	-0.012 (-0.33)	-0.012 (-0.52)	-0.015 (-0.67)
No. of Business Segments	-0.001 (-0.52)	-0.002 (-0.58)	0.000 (0.21)	-0.023** (-1.88)	-0.020* (-1.65)	-0.000 (-0.08)	-0.000 (-0.07)	0.000 (0.09)
Free Cash Flow	0.050 (1.30)	0.022 (0.47)	0.053 (1.46)	0.059 (0.29)	0.148 (0.80)	-0.001 (-0.04)	-0.015 (-0.33)	0.003 (0.07)
Young IPO Firm	0.000 (0.01)	-0.007 (-0.67)	-0.004 (-0.46)	-0.032 (-0.55)	-0.012 (-0.22)	0.030*** (2.88)	0.024** (2.06)	0.029*** (2.78)
Fraction of Foreign M&A	0.049** (2.05)	0.062** (2.33)	0.046** (2.08)					
Dummy for B/H Share				-0.098 (-1.43)	-0.114* (-1.92)			
Year FE	No	No	No	Yes	Yes	No	No	No
Industry FE	Yes	Yes	Yes	Yes	Yes	No	No	No
Partial R^2				0.01				0.014
Cragg-Donald Wald F -statistic				9.25				20.36
5% maximal IV relative bias				16.85				16.85
10% maximal IV relative bias				10.27				10.27
No. of obs.	4,094	4,094	4,094	355	355	5,917	5,917	5,917
R^2	0.044	0.036	0.200	0.128	0.251	0.031	0.021	0.052

Table XI, part 1

Table XI, part 2

7.12 Table XII: Corporate Governance

Read:

- The table studies market reaction to foreign M&A, earnings management, CEO turnover, and pay-for-performance sensitivity.
- Foreign-experienced directors are associated with lower earnings management and stronger governance mechanisms.
- CEO turnover and compensation become more sensitive to performance in relevant specifications.

Detailed interpretation: The table shows that foreign-experienced directors affect monitoring, not just advising. This is important because boards have both advisory and disciplinary roles.

Economic message: Returnee directors can import governance norms and monitoring practices from stronger institutional environments.

Table XII
Corporate Governance

This table relates corporate governance to the proportion of directors with foreign experience. The dependent variable in columns 1 and 2 is the firm's cumulative abnormal return starting two days before the announcement of a foreign merger and acquisition up to five days after the announcement. The dependent variable in columns 3 to 5 is a proxy increasing in the extent of earnings management computed as in Kothari, Leone, and Wasley (2005). The dependent variable in columns 6 and 7 is a dummy that takes a value of one if there is a CEO turnover event, and zero otherwise. The dependent variable in columns 8 and 9 is the natural logarithm of the average cash and bonus pay of the top three executives. "High CG Ranking" is a dummy equal to one if at least one director obtained his/her foreign experience from a country with the highest La Porta et al. (1998) anticorruption rights index, and zero otherwise. Except for column 4, estimates are obtained by ordinary least squares; in column 4, we report instrumental variable estimates. The instrumental variables include "Provincial Policy," a dummy that takes a value of one in years following the implementation of the policy in each province, and interaction variables between the policy dummy and firm ownership characteristics "State," "Foreign Ownership," and "Block" in 1999 (the beginning of the sample period). If a firm enters our sample later than 1999, firm ownership characteristics computed in the year of the firm's entry in the sample are used to construct the interaction terms. Variable definitions are in the Appendix. *t*-statistics computed with robust standard errors clustered at the firm level are reported in parentheses. All models include a constant, but the coefficients are not reported. We report the partial R^2 of the instruments in the first stage and the Cragg-Donald Wald F -statistics. We also report the Stock-Yogo weak ID test critical values for the Cragg-Donald Wald F -statistics. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	CAR		Earnings Management			CEO Turnover		Pay for Performance Sensitivity	
	OLS	OLS	OLS	IV	OLS	OLS	OLS	OLS	OLS
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Foreign Experience	0.060 (0.71)	-0.350 (-1.53)	-0.008 (-0.92)	-0.150** (-2.22)	0.028 (1.13)	0.019 (0.59)	-0.078 (-0.94)	0.889*** (6.17)	1.080*** (3.94)
Foreign Experience × High CG Ranking		0.411* (1.83)		-0.040* (-1.66)	0.105 (1.29)		0.105 (1.29)	-0.225 (-0.84)	
Foreign Experience × ROA					-0.387 (-1.14)	1.208 (1.20)	1.782 (1.17)	-4.215 (-1.56)	
Foreign Experience × ROA × High CG Ranking						-0.452*** (-7.07)	-0.462*** (-7.19)	2.696*** (10.40)	2.736*** (10.52)
Foreign Ownership	-0.030 (-1.53)	-0.033 (-1.65)	-0.004 (-1.54)	0.007 (1.28)	-0.004 (-1.48)	-0.011 (-1.33)	-0.012 (-1.38)	0.153*** (4.34)	0.153*** (4.37)

(Continued)

Table XII, part 1

Table XII—Continued

	CAR		Earnings Management			CEO Turnover		Pay for Performance Sensitivity	
	OLS	OLS	OLS	IV	OLS	OLS	OLS	OLS	OLS
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Block	-0.006 (-0.13)	0.003 (0.06)	-0.003 (-0.51)	-0.008 (-1.23)	-0.003 (-0.54)	0.030 (1.60)	0.029 (1.56)	-0.555*** (-5.46)	-0.551*** (-5.42)
State	-0.012 (-0.63)	-0.011 (-0.60)	0.000 (0.02)	-0.003 (-1.17)	0.000 (0.06)	-0.017** (-2.38)	-0.017** (-2.41)	-0.130*** (-4.02)	-0.130*** (-4.01)
Size	0.003 (0.29)	0.002 (0.21)	0.003*** (2.51)	0.004*** (3.08)	0.003** (2.56)	-0.009*** (-2.62)	-0.009*** (-2.61)	0.308*** (17.60)	0.307*** (17.54)
Leverage	0.046 (0.84)	0.047 (0.88)	0.011** (2.42)	0.011** (2.25)	0.011** (2.44)	0.013 (0.78)	0.012 (0.72)	0.032 (0.46)	0.037 (0.53)
No. of Business Segments	0.005 (0.69)	0.005 (0.75)	0.002*** (3.23)	0.002*** (3.16)	0.002*** (3.25)	-0.004* (-1.73)	-0.004* (-1.70)	0.028*** (2.65)	0.029*** (2.63)
Free Cash Flow	0.140 (0.85)	0.155 (0.94)	0.131*** (10.48)	0.133*** (10.45)	0.131*** (10.48)	0.086* (1.89)	0.084* (1.84)	-0.638*** (-4.23)	-0.639*** (-4.19)
Young IPO Firm	0.011 (0.50)	0.008 (0.37)	0.010*** (4.78)	0.010*** (4.81)	0.010*** (4.79)	-0.017** (-2.52)	-0.017** (-2.56)	0.165*** (5.57)	0.166*** (5.61)
Stock Volatility								-0.300 (-0.70)	-0.301 (-0.70)
CEO Age						0.003*** (6.85)	0.003*** (6.66)		
CEO Tenure						-0.005*** (-3.16)	-0.004*** (-3.15)		
Year FE	No	No	No	No	No	Yes	Yes	Yes	Yes
Industry FE	No	No	No	No	No	Yes	Yes	Yes	Yes
Partial R^2									
Cragg-Donald Wald F -statistic									
5% maximal IV relative bias									
10% maximal IV relative bias									
No. of obs.	185	185	13,084	13,084	13,084	13,615	13,615	10,233	10,233
R^2	0.036	0.052	0.019	0.020	0.019	0.094	0.094	0.369	0.369

1670

The Journal of Finance

Table XII, part 2

7.13 Table XIII: Management Practices

Read:

- The table studies asset turnover and post-M&A performance.
- It interacts foreign experience with management-practice rankings of the countries where directors obtained experience.
- Directors exposed to countries with stronger management practices are associated with better operating efficiency.

Detailed interpretation: This table provides mechanism evidence that knowledge about management practices is part of the effect. The country-level ranking interaction connects the content of director experience to the type of corporate improvement observed.

Economic message: Foreign experience is not homogeneous; experience from high-management-practice countries helps firms improve operating efficiency.

Table XIII
Management Practices

This table relates operating performance to the proportion of directors with foreign experience. The dependent variable in columns 1 to 3 is "Asset Turnover," from which we subtract the year median. The dependent variable in columns 4 and 5 is a firm's return over the 24 months after the announcement of an international merger and acquisition minus the market return during the same period. "High MP Ranking" is a dummy that equals one if at least one director obtained his/her foreign experience from one of the top three countries according to Bloom et al.'s (2012) monitoring management score, and zero otherwise. Except for column 2, estimates are obtained by ordinary least squares; in column 2, we report instrumental variable estimates. The instrumental variables include "Provincial Policy," a dummy that takes a value of one in years following the implementation of the policy in each province, and interaction variables between the policy dummy and firm ownership characteristics "State," "Foreign Ownership," and "Block" in 1999 (the beginning of the sample period). If a firm enters our sample later than 1999, firm ownership characteristics computed in the year of the firm's entry in the sample are used to construct the interaction terms. Variable definitions are in the Appendix. *t*-statistics computed with robust standard errors clustered at the firm level are reported in parentheses. All models include a constant, but the coefficients are not reported. We report the partial R^2 of the instruments in the first stage and the Cragg-Donald Wald F -statistics. We also report the Stock-Yogo weak ID test critical values for the Cragg-Donald Wald F -statistics. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	Asset Turnover			Post M&A Performance	
	OLS (1)	IV (2)	OLS (3)	OLS (4)	OLS (5)
Foreign Experience	0.261*** (3.51)	1.045* (1.85)	0.131 (1.26)	0.193 (0.44)	-0.614 (-1.32)
Foreign Experience × High MP Ranking			0.171* (1.67)		0.914** (2.16)
Foreign Ownership	0.101 (0.84)	-0.288 (-1.06)	0.105 (0.88)	-0.050 (-0.41)	-0.035 (-0.29)
Block	0.194*** (3.45)	0.212*** (3.64)	0.194*** (3.47)	-0.214 (-0.80)	-0.221 (-0.81)
State	0.065*** (3.50)	0.077*** (3.55)	0.064*** (3.48)	-0.198* (-1.66)	-0.201* (-1.69)
Size	0.044*** (4.84)	0.036*** (3.09)	0.043*** (4.77)	-0.007 (-0.15)	-0.008 (-0.17)
Leverage	0.160*** (4.40)	0.160*** (4.34)	0.160*** (4.39)	-0.393 (-1.57)	-0.398 (-1.59)
No. of Business Segments	-0.001 (-0.18)	-0.002 (-0.38)	-0.001 (-0.17)	-0.056* (-1.81)	-0.048 (-1.53)
Free Cash Flow	0.723*** (11.59)	0.719*** (11.47)	0.723*** (11.57)	1.352** (2.06)	1.313** (2.01)
Young IPO Firm	0.023* (1.93)	0.025** (2.03)	0.023* (1.90)	-0.085 (-0.85)	-0.081 (-0.82)
Stock Volatility	-0.051 (-0.13)	-0.105 (-0.26)	-0.032 (-0.08)	4.415 (1.56)	4.238 (1.49)
Sales Growth	0.184*** (17.49)	0.180*** (16.16)	0.185*** (17.52)		
Partial R^2		0.009			

(Continued)

1674

The Journal of Finance[®]

Table XIII—Continued

	Asset Turnover			Post M&A Performance	
	OLS (1)	IV (2)	OLS (3)	OLS (4)	OLS (5)
Cragg-Donald Wald F -statistic		31.122			
5% maximal IV relative bias		16.85			
10% maximal IV relative bias		10.27			
Industry FE	Yes	Yes	Yes	No	No
No. of obs.	13,183	13,183	13,183	233	233
R^2	0.247	0.217	0.248	0.097	0.107

Table XIII, part 1

Table XIII, part 2

7.14 Table XIV: Corporate Governance, Management Practices, and Corporate Valuations

Read:

- The table interacts foreign experience with rankings for corporate governance and management-practice quality in directors' experience countries.
- The valuation benefits are stronger when directors' foreign experience comes from countries with high-quality governance or management practices.
- Controls remain similar to earlier performance specifications.

Detailed interpretation: Table XIV ties the mechanism evidence back to firm value. It shows that market valuations rise more when the director's foreign experience is plausibly more relevant for governance or management improvements.

Economic message: The market values not just foreignness, but the content and institutional quality of the director's foreign experience.

Table XIV—Continued

	(1)	(2)	(3)
No. of Business Segments	−0.027*** (−2.67)	−0.027*** (−2.73)	−0.027*** (−2.70)
Free Cash Flow	0.957*** (5.86)	0.959*** (5.87)	0.957*** (5.86)
Young IPO Firm	−0.130*** (−4.33)	−0.130*** (−4.31)	−0.131*** (−4.34)
Stock Volatility	5.818*** (11.08)	5.812*** (11.06)	5.823*** (11.10)
No. of obs.	13,722	13,722	13,722
R ²	0.215	0.215	0.215

Table XIV
Corporate Governance, Management Practices, and Corporate Valuations

This table presents the ordinary least squares estimates relating firm value to the proportion of directors with foreign experience. The dependent variable is the firm's MTB, from which we subtract the industry-year median. "High CG Ranking" is a dummy that equals one if at least one director obtained his/her foreign experience from a country with the highest La Porta et al. (1998) antidirector rights index, and zero otherwise. "High MP Ranking" is a dummy that equals one if at least one director obtained his/her foreign experience from one of the top three countries according to Bloom et al.'s (2012) monitoring management score, and zero otherwise. Variable definitions are in the Appendix. *t*-statistics computed with robust standard errors clustered at the firm level are reported in parentheses. All models include a constant, but the coefficients are not reported. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)
Foreign Experience	0.425** (2.06)	0.162 (0.55)	0.046 (0.15)
Foreign Experience × High MP Ranking	0.544** (2.57)		0.411* (1.79)
Foreign Experience × High CG Ranking		0.743** (2.54)	0.529* (1.66)
Foreign Ownership	0.339 (1.48)	0.304 (1.32)	0.318 (1.39)
Block	0.548*** (5.49)	0.550*** (5.49)	0.550*** (5.51)
State	−0.130*** (−3.57)	−0.131*** (−3.57)	−0.131*** (−3.59)
Size	−0.503*** (−22.97)	−0.502*** (−22.93)	−0.503*** (−23.00)
Leverage	0.223** (1.97)	0.222** (1.97)	0.222** (1.96)

(Continued)

VI. Conclusions

The brain drain from emerging markets may have not only costs, but also positive, indirect, benefits. Talented individuals migrating to foreign countries accumulate knowledge and skills. The experiences that they accumulate, coupled with their presumably higher ex ante ability, may allow their talent to flourish. If these highly skilled emigrants decide to return, the experience they gained abroad can benefit their home country in which case the brain drain becomes a brain gain.

This paper documents a specific channel leading to brain gain. We show that, when individuals with foreign experience join corporate boards, firm performance improves and firms are run differently. The positive effects on firm valuations are large relative to the compensation of board members, which is moderate in China. Thus, our results suggest that, by successfully competing in attracting talent, governments can greatly benefit firm productivity and performance.

While our results suggest that directors with foreign experience can facilitate the transfer of knowledge and provide connections in foreign countries, it remains to be explored to what extent similar effects could be achieved through alternative channels such as foreign ownership. Our results suggest that directors with foreign experience are somewhat special. A possibility is that the background they share with the locals allows them to overcome cultural barriers. We consider this an exciting area for future research.

Initial submission: October 30, 2012; Final version received: May 8, 2014
Editor: Michael Roberts

Table XIV, part 1

Table XIV, part 2

8 Result map

- **Institutional shock.** Provincial policies to attract skilled emigrants provide staggered variation in local supply.
- **First stage.** Policy adoption increases the share of foreign-experienced directors on local boards.
- **Main effect.** Foreign-experienced directors raise valuation, productivity, and profitability.
- **Robustness.** Effects survive controls for ownership, board structure, director ability, province trends, and alternative samples.
- **Internationalization channel.** Directors help firms do foreign M&A, foreign private placements, and foreign sales, especially in countries connected to the directors' experience.
- **Governance channel.** Directors reduce earnings management and improve CEO discipline and incentives.
- **Management-practice channel.** Directors from high-management-practice countries are associated with better operating efficiency.
- **Content of experience.** Benefits depend on the type and country of foreign experience, suggesting that the mechanism is knowledge transfer rather than only innate ability.

9 Short summary

- The paper studies whether directors with foreign education or work experience improve firm outcomes in China.
- It uses a hand-collected director biography data set and Chinese listed-firm data from 1999 to 2009.
- The main identification strategy exploits staggered provincial policies to attract skilled emigrants.
- The director labor market is local, so provincial policies plausibly shift the supply of foreign-experienced directors for local firms.
- Firms hiring directors with foreign experience experience higher valuations, TFP, and profitability.
- The effects appear in firm policies that boards influence: foreign acquisitions, foreign placements, foreign sales, earnings management, CEO turnover, CEO pay sensitivity, and management efficiency.
- The paper provides suggestive evidence that the content of foreign experience matters beyond innate ability.
- The main contribution is to show a concrete corporate-governance channel through which migration can produce brain gain in emerging markets.

10 Conclusion

10.1 Core conclusion

Giannetti, Liao, and Yu show that foreign-experienced directors can be valuable board human capital in emerging markets. The addition of such directors is associated with better valuation, productivity, and profitability, and the IV evidence suggests that at least part of the relationship is causal for firms whose boards respond to policy-induced changes in local talent supply.

10.2 Economic intuition

Foreign experience matters because it combines three assets: knowledge, networks, and monitoring norms. Directors with experience abroad may understand global management practices, have connections to foreign capital and product markets, and be less tied to local political or business networks that weaken monitoring incentives. These assets are especially valuable in China, where firms face talent shortages and governance frictions.

10.3 Incremental contribution revisited

The paper's incremental contribution is to turn the abstract idea of brain gain into a measurable corporate mechanism. Rather than asking whether return migration raises aggregate human capital in general, it shows that returnee and foreign-trained individuals can affect firms through board seats, where they have both advisory and monitoring authority. This makes the board a channel for international knowledge transmission.

10.4 Why the evidence is persuasive

The design has three mutually reinforcing elements. First, local policies create plausibly exogenous variation in the supply of foreign-experienced directors. Second, the paper documents that director labor markets are local, so provincial policies should affect local firms. Third, the observed outcomes match the mechanisms: governance improves, internationalization rises, and management-practice-related efficiency improves when the director's country of experience is relevant.

10.5 Limitations

The paper cannot fully separate innate ability from experience because the people who acquire foreign experience may also be unusually talented. The policy instruments identify effects for firms whose board composition responds to supply shocks, not necessarily all firms. Some mechanism tests are suggestive because the type and country of foreign experience are not randomly assigned.

10.6 Takeaway

The paper's message is that international human capital can reshape firms even without foreign ownership. When skilled emigrants return or reconnect with domestic firms as directors, they can transfer knowledge, connections, and governance norms. In this sense, brain drain can become brain gain through corporate boards.